Project Report

Project Report Format

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Contents

1. INTRODUCTION 2

1.1 Project Overview . . . 2

1.2 Purpose . . . 2

2. IDEATION PHASE 2

2.1 Problem Statement . . . 2

2.2 Empathy Map Canvas . . . 2

2.3 Brainstorming . . . 2

3. REQUIREMENT ANALYSIS 3

3.1 Customer Journey map . . . 3

3.2 Solution Requirement . . . 3

3.3 Data Flow Diagram . . . 4

3.4 Technology Stack . . . 5

4. PROJECT DESIGN 5

4.1 Problem Solution Fit . . . 5

4.2 Proposed Solution . . . 6

4.3 Solution Architecture . . . 6

5. PROJECT PLANNING & SCHEDULING 7

5.1 Project Planning . . . 7

6. FUNCTIONAL AND PERFORMANCE TESTING 8

6.1 Performance Testing . . . 8

7. RESULTS 9

7.1 Output Screenshots . . . 9

8. ADVANTAGES & DISADVANTAGES 12

9. CONCLUSION 12

10. FUTURE SCOPE 13

11. APPENDIX 13

1. INTRODUCTION

1.1 Project Overview

Global Olympic Sports Performance Analysis uses Snowflake and Tableau to turn an Olympic dataset (1992–2026, 8,988 records, 56 columns) into an interactive dashboard and story. It shows medal performance by country, medal type, medal points, sports participation and athlete events. The data is loaded into Snowflake with SQL, connected to Tableau, summarised in 7 worksheets and combined into Dashboard 1 and Story 1.

1.2 Purpose

The purpose of the project is to identify leading countries, understand participation across 50 sports, highlight the most active athletes and give analysts, federations, journalists and fans a reusable dashboard that replaces slow manual comparison of raw data.

2. IDEATION PHASE

2.1 Problem Statement

The customer problem statement describes the customer's point of view so that the team can find the ideal solution.

I am	A sports performance analyst / Olympic data officer who studies country, sport and athlete results from 1992 to 2026
I'm trying to	Understand which countries, sports and athletes drive Olympic performance
But	The data sits in 8,988 raw rows with four mixed record types and cannot be compared quickly
Because	There is no single visual view that totals medals, ranks countries and counts athletes per sport
Which makes me feel	Overwhelmed and unsure, unable to explain results to stakeholders

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Sports analyst	Compare medal performance across countries	Raw rows are hard to rank and compare	Medals are split into gold, silver and bronze columns	Confused and slow to report
PS-2	Sports federation planner	Find which sports attract the most athletes	Participation is not summarised by sport	50 sports and 8,000 athlete records need aggregation	Unable to plan funding
PS-3	Sports journalist / student	Tell the story of dominant nations and top athletes	No ready narrative from raw tables	There is no dashboard or story flow	Frustrated, loses audience interest

2.2 Empathy Map Canvas

The empathy map captures what the persona, a Sports Performance Analyst, says, thinks, does and feels.

SAYS • "Which country is really leading?" • "Show me the top 10 quickly." • "How many athletes per sport?"	**PERSONA** **Sports Performance Analyst** *(Role / User Type)*	**THINKS** • The raw CSV is too big to read. • Rankings must be accurate. • One clear chart saves hours.
DOES • Sorts and filters sheets by hand. • Copies numbers into slides. • Compares gold, silver, bronze.		**FEELS** • Frustrated by manual work. • Anxious about errors. • Excited by interactive visuals.

PAINS	GAINS
Time-consuming manual aggregation; hard to rank 60 countries and 50 sports; risk of double counting athletes; static tables without interactivity	Instant top-10 rankings; colour-coded gold / silver / bronze comparison; one dashboard and a guided story; faster evidence-based decisions

2.3 Brainstorming

Each team member listed ideas without judging them (Step 1). The ideas were then grouped, rated on feasibility and importance, and the final ideas were selected (Step 2).

S.No	Team Member	Idea / Suggestion	Category	Group
1	Team Member 1	Bar chart of top 10 countries by total medals	Country performance	1
2	Team Member 2	Stacked bar comparing gold, silver and bronze by country	Country performance	1
3	Team Member 3	Sports participation by number of athletes	Sport and athlete insight	2
4	Team Member 4	Top 10 athletes by number of events	Sport and athlete insight	2
5	Team Member 1	Ranking of countries by medal points	Country performance	1
6	Team Member 2	Calculated field for total medals	Presentation / preparation	3
7	Team Member 3	Filters on country, sport and dataset type	Presentation / preparation	3
8	Team Member 4	Dashboard combining all views and a story to explain it	Presentation / preparation	3

Group	Final Idea	Feasibility	Importance	Priority	Selected
1	Country performance views: total medals, gold/silver/bronze comparison, medal points	High	High	1	Yes
2	Sports and athlete views: participation, top 10 sports, top athletes	High	Medium	2	Yes
3	Dashboard, story, filters and calculated field	Medium	High	1	Yes
4	Efficiency analysis and host-city trend (future work)	Medium	Low	3	No

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The journey map shows the customer's experience stage by stage.

Phase of Journey	Stage 1: Explore the data	Stage 2: Build the visuals	Stage 3: Share the insights
Actions	Opens athelete_events.csv; reviews 4 dataset types; identifies key fields	Connects data to Tableau; creates 7 worksheets; applies filters	Assembles Dashboard 1; builds Story 1; presents findings
Touchpoint	CSV file; Snowflake table; Tableau data source tab	Tableau worksheets; filter and measure shelves; calculated field	Dashboard canvas; story points; .twbx / Tableau Public
Customer Thought	"Too much raw data"; "Which measures matter?"	"Is this chart answering my question?"; "Are athletes counted once?"	"Can others understand this?"; "I can reuse this"
Customer Feeling	Overwhelmed, curious	Focused, increasingly confident	Satisfied, proud
Process Ownership	Data lead (Team Member 1)	Visualization leads (Team Members 2, 3)	Dashboard and story lead (Team Member 4)
Opportunities	Single clean source; clear KPI definitions	Reusable worksheet templates; interactive filters	Add a map view; publish to Tableau Public

3.2 Solution Requirement

Functional requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	Data Access & Integration	Load athelete_events.csv (8,988 rows, 4 dataset types) into Snowflake and connect it to Tableau	High
2	Data Processing	Filter by dataset_type, set geographic role on country, create calculated field Total Medals Calculated	High
3	Business Analysis	Total medals, gold / silver / bronze split, medal points, distinct athlete counts per sport and per name	High

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
4	Business Insights & Analytical Outputs	Seven worksheets, Dashboard 1 and Story 1 showing leaders, participation and top athletes	High
5	User Interaction & Analysis	Filters on country, sport, name and dataset type to compare and drill into results	Medium
6	Data Update & Reporting	Refresh the extract when the CSV changes; export the workbook (.twbx) and PDF documentation	Medium

Non-functional requirements

NFR No.	Requirement	Description
NFR-1	Usability	Clear titles, colour legend and few clicks to reach any insight; usable by non-technical users
NFR-2	Security	No sensitive personal data; access to Snowflake and workbooks limited to the team
NFR-3	Reliability	Packaged workbook (.twbx) keeps data with visuals; medal totals are consistent across views
NFR-4	Performance	Dashboard renders quickly on 8,988 rows using an extract; queries return in seconds
NFR-5	Availability	Workbook available through the packaged file or Tableau Public
NFR-6	Scalability	Supports additional Games, countries and sports without redesigning the views

User stories

User Type	Epic	Story	User Story / Task	Priority	Release
Analyst	Country performance	USN-1	See the top 10 countries by total medals	High	Sprint-1
Analyst	Country performance	USN-2	Compare gold, silver and bronze by country	High	Sprint-1
Analyst	Medal points	USN-3	Rank countries by medal points	Medium	Sprint-2
Federation planner	Sports participation	USN-4	See the number of athletes in each sport	High	Sprint-2
Journalist	Top athletes	USN-5	See the top 10 athletes by events	Medium	Sprint-2
All users	Dashboard and story	USN-6	Explore all views in one dashboard and follow a story	High	Sprint-3

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) shows how data enters and leaves the system, what changes the information and where data is stored.

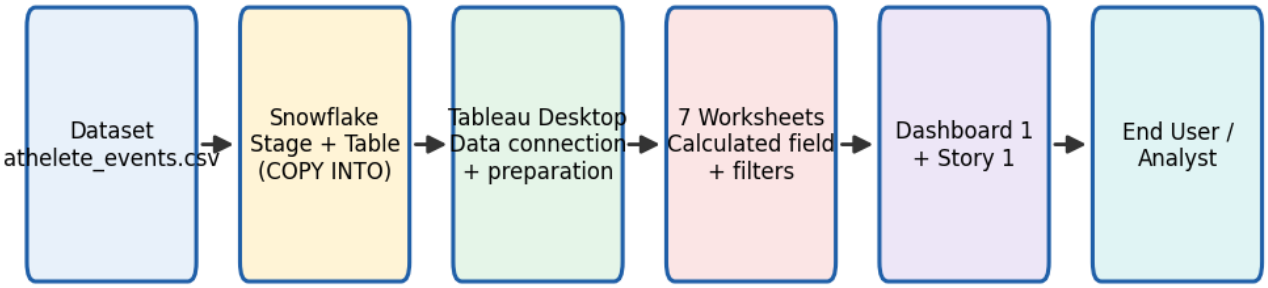


Figure: DFD Level 0 – data flow from the dataset to the end user

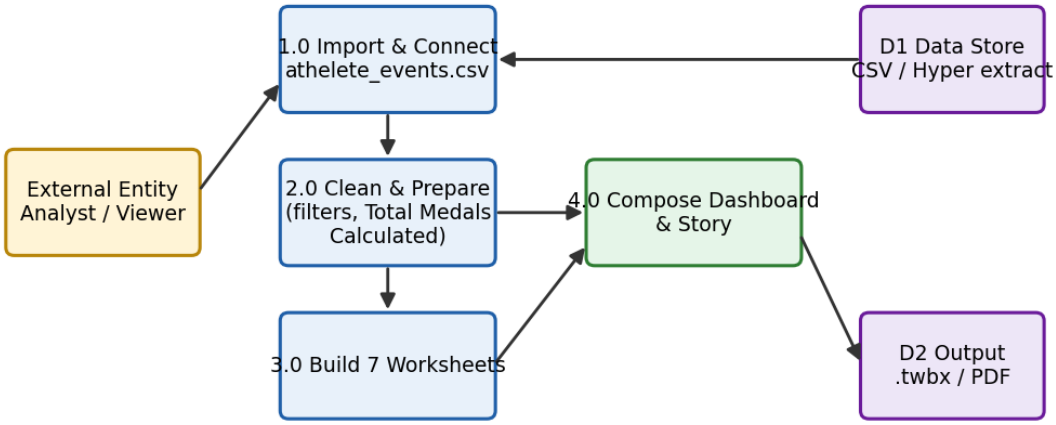


Figure: DFD Level 1 – processes inside the Tableau workflow

3.4 Technology Stack

S.No	Component	Technology	Justification / Purpose
1	Programming / Query Language	SQL (Snowflake), Tableau calculated fields	SQL verifies totals; calculated field creates Total Medals
2	Database / Data Warehouse	Snowflake / CSV	Holds 8,988 records in one table with a record-type column
3	Data Ingestion	Snowflake Stage / COPY INTO / Tableau Data Connections	Loads athelete_events.csv and connects it to Tableau
4	Data Transformation and Business Logic	Tableau Desktop, Snowflake SQL	Filters by dataset_type, COUNTD of athletes, total medals
5	Data Visualization / Analytics	Tableau (bar and stacked bar charts)	Seven worksheets, dashboard and story
6	Front-End / User Interface	Tableau Public / Tableau Desktop	Interactive filters and story navigation for end users

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit shows that the solution actually solves the customer's problem and fits the customer's behaviour.



Figure: Problem–Solution Fit canvas for Global Olympic Sports Performance Analysis

4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Olympic performance data (1992–2026) is large and mixed, which makes it hard to compare countries, sports and athletes quickly.
2	Idea / Solution Description	Load the dataset into Snowflake, connect it to Tableau and build 7 worksheets, a dashboard and a story that reveal medal leaders, participation by sport and top athletes.
3	Novelty / Uniqueness	One consistent view combining medal counts, medal points, sports participation and athlete event counts across 18 Games.
4	Social Impact / Customer Satisfaction	Helps committees, fans and students understand Olympic performance and supports data-driven sports development.
5	Business Model (Revenue Model)	Free educational dashboard; extendable to sponsor analytics and federation consulting reports.
6	Scalability of the Solution	Add new Games, sports and countries; connect Tableau live to Snowflake for automatic refresh.

- **Objective:** identify leading countries, sports participation and top athletes using interactive visuals.
- **Scope:** 60 countries, 50 sports, 8,000 athlete records, 910 country-by-Games medal records, 18 Games.
- **Resources:** athelete_events.csv, Snowflake account, Tableau Desktop, SmartBridge templates.
- **Expected outcome:** 7 worksheets, Dashboard 1, Story 1 and phase-wise PDF documentation.

4.3 Solution Architecture

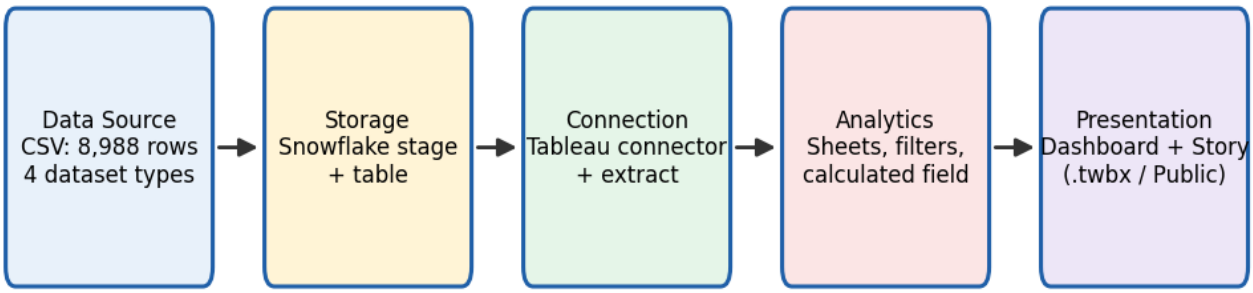


Figure: Solution architecture (data source to dashboard and story)

Layer	Component	Description
Data source	athelete_events.csv	8,988 rows × 56 columns with dataset_type values athletes, medal_results, alltime_country_rankings and games_summary
Storage	Snowflake stage and table	CSV is staged and loaded with COPY INTO into table OLYMPIC_EVENTS
Connection	Tableau connector / extract	Connects to the table or CSV; extract improves performance
Analytics	Worksheets, filters, calculated field	Total Medals Calculated = total_gold + total_silver + total_bronze; COUNTD of athlete_id
Presentation	Dashboard 1 and Story 1	Interactive dashboard and guided story
Delivery	.twbx / Tableau Public / PDF	Packaged workbooks and documentation

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was split into four sprints and estimated with story points (39 in total).

Sprint	Functional Requirement (Epic)	Story	User Story / Task	Points	Priority	Team
Sprint-1	Data Access & Integration	USN-1	Collect athelete_events.csv and study its four dataset types	3	High	Member 1, 2
Sprint-1	Data Access & Integration	USN-2	Load the CSV into Snowflake (stage, table, COPY INTO) and connect to Tableau	2	High	Member 1
Sprint-1	Data Processing	USN-3	Filter dataset types, create Total Medals Calculated, set geographic role	5	High	Member 2, 3
Sprint-2	Business Insights	USN-4	Worksheets 1–3: medal distribution, gold/silver/bronze, sports participation	8	High	Member 3, 4
Sprint-2	Business Insights	USN-5	Worksheets 4–7: top athletes, medal points, top countries, top sports	8	High	Member 1, 4
Sprint-3	User Interaction	USN-6	Build Dashboard 1 with layout and filters	5	High	Member 2, 3
Sprint-3	User Interaction	USN-7	Create Story 1 with captions and flow	3	Medium	Member 4
Sprint-4	Data Update & Reporting	USN-8	Performance testing, results, documentation and demo	5	Medium	All members

Sprint	Total Points	Duration	Start Date	End Date (Planned)	Points Completed	Release Date (Actual)
Sprint-1	10	6 Days	12 Oct 2026	17 Oct 2026	10	17 Oct 2026
Sprint-2	16	6 Days	18 Oct 2026	23 Oct 2026	16	23 Oct 2026
Sprint-3	8	6 Days	24 Oct 2026	29 Oct 2026	8	29 Oct 2026
Sprint-4	5	3 Days	30 Oct 2026	01 Nov 2026	5	01 Nov 2026

Velocity: average velocity = 39 story points / 4 sprints = 9.75 story points per sprint. All story points were completed by 01-11-2026.

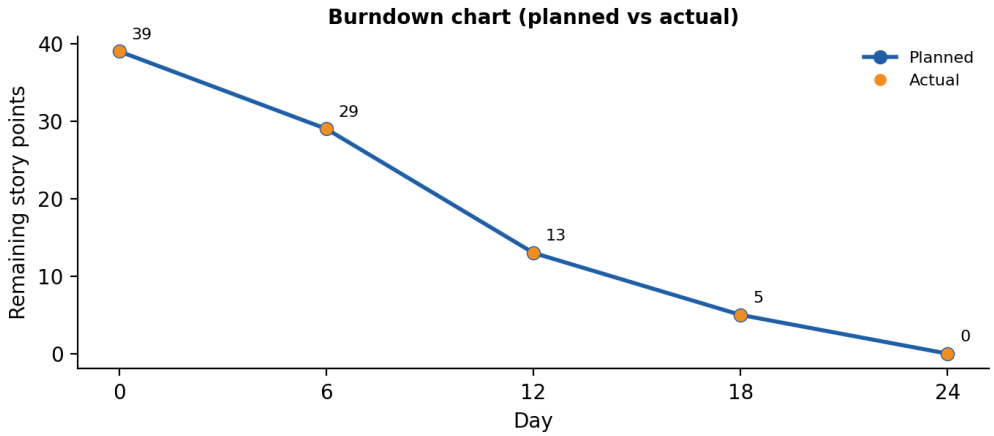


Figure: Burndown chart – remaining story points (planned and actual)

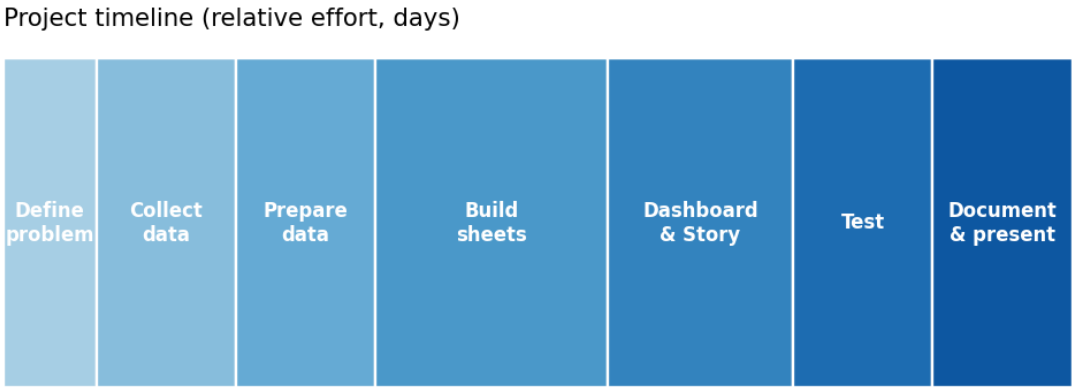


Figure: Project timeline (relative effort)

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Snowflake

S.No.	Parameter	Values
1	Data rendered / loaded	8,988 rows × 56 columns loaded into OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS (athletes 8,000; medal_results 910; alltime_country_rankings 60; games_summary 18)
2	Warehouse used	OLYMPIC_WH, size XSMALL, auto-suspend 60 seconds
3	Data preprocessing	File format CSV_FF (skip header, treat blanks as NULL); dataset_type filter in every query
4	Number of analysis queries	5 aggregate queries (total medals, gold/silver/bronze, athletes per sport, athletes per name, medal points) plus 1 load check
5	Query features used	GROUP BY, ORDER BY, LIMIT, SUM, COUNT(DISTINCT), WHERE on dataset_type
6	Result validation	All query results match the Tableau worksheets (e.g. United States 735 medals, Surfing 201 athletes, USA 1,443 medal points)
7	Connection to Tableau	Snowflake connector with extract; dashboard refresh uses the table OLYMPIC_EVENTS

Test ID	Scenario	Expected result	Status
SF-01	COPY INTO load	8,988 rows loaded without errors	Pass
SF-02	Count by dataset_type	8,000 / 910 / 60 / 18	Pass
SF-03	Top 10 countries by medals	United States 735, China 696, Russia 429	Pass

Test ID	Scenario	Expected result	Status
SF-04	Athletes per sport	Surfing 201, Canoe/Kayak 199, Handball 193	Pass
SF-05	Medal points ranking	United States 1,443, China 1,386	Pass

Tableau

S.No.	Parameter	Values
1	Data rendered	8,988 rows × 56 columns (4 dataset types)
2	Data preprocessing	dataset_type filters, geographic role on country, extract (.hyper)
3	Utilization of data filters	12 filter instances across 7 worksheets on 5 fields: Country, Dataset Type, Measure Names, Sport, Name
4	No. of calculation fields	1 – Total Medals Calculated = [total_gold] + [total_silver] + [total_bronze]
5	No. of visualizations / graphs	7 worksheets (bar charts and one stacked bar)
6	Dashboard design	1 dashboard (Dashboard 1) with interactive filters, titles and legend
7	Story design	1 story (Story 1) with the caption “Global Olympic Sports Performance Analysis”
8	Export / packaging	2 packaged workbooks (.twbx): Dashboard and Story

Test ID	Scenario	Expected result	Actual result	Status
TC-01	Load CSV in Tableau	56 fields visible	Connected	Pass
TC-02	Total Medals Calculated	gold + silver + bronze	USA 735 (250 + 208 + 277)	Pass
TC-03	Top 10 countries by total medals	United States first	USA 735, China 696, Russia 429	Pass
TC-04	Sports participation (COUNTD)	Distinct athletes per sport	Surfing 201 to Luge 100	Pass
TC-05	Top 10 athletes	10 names shown	Max 17 records (Wei Ngozi)	Pass
TC-06	Medal points	Descending order	USA 1,443; China 1,386	Pass
TC-07	Dashboard filters	Linked views update	Views update	Pass
TC-08	Story navigation	Story point opens	Dashboard 1 displayed	Pass

7. RESULTS

7.1 Output Screenshots

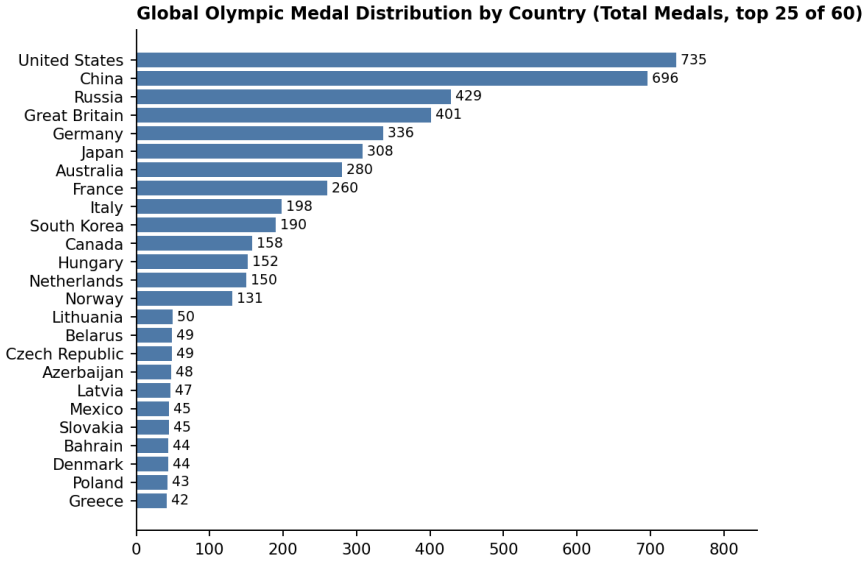


Figure: Worksheet 1: Global Olympic Medal Distribution by Country

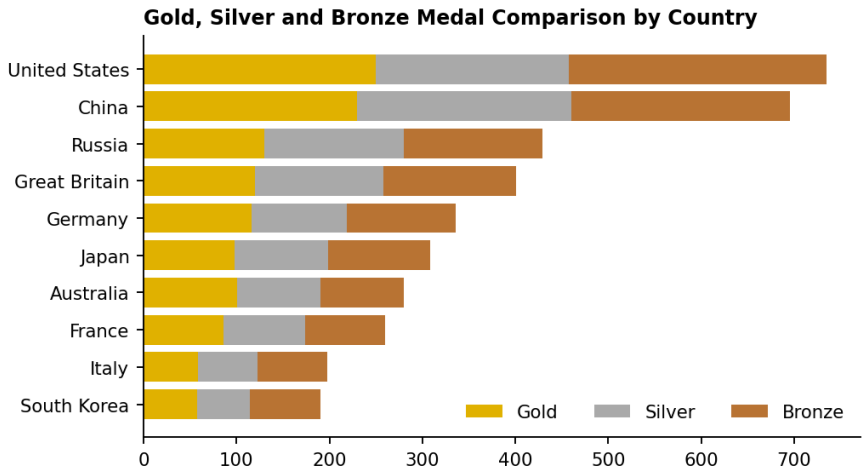


Figure: Worksheet 2: Gold, Silver and Bronze Medal Comparison by Country

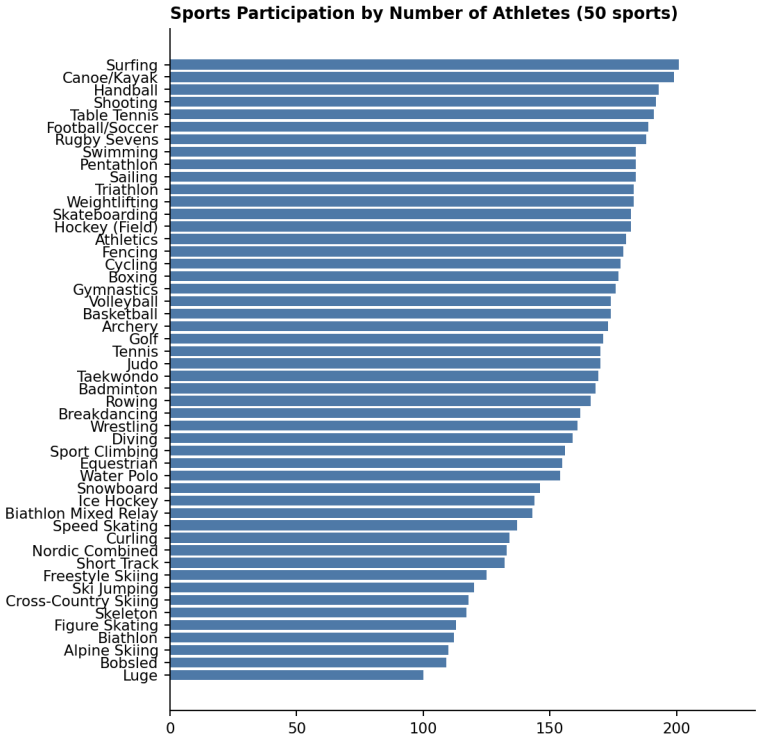


Figure: Worksheet 3: Sports Participation by Number of Athletes

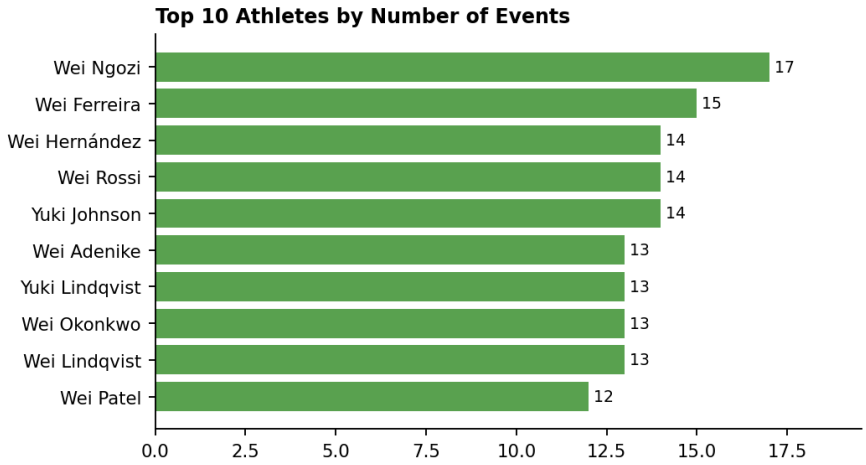


Figure: Worksheet 4: Top 10 Athletes by Number of Events

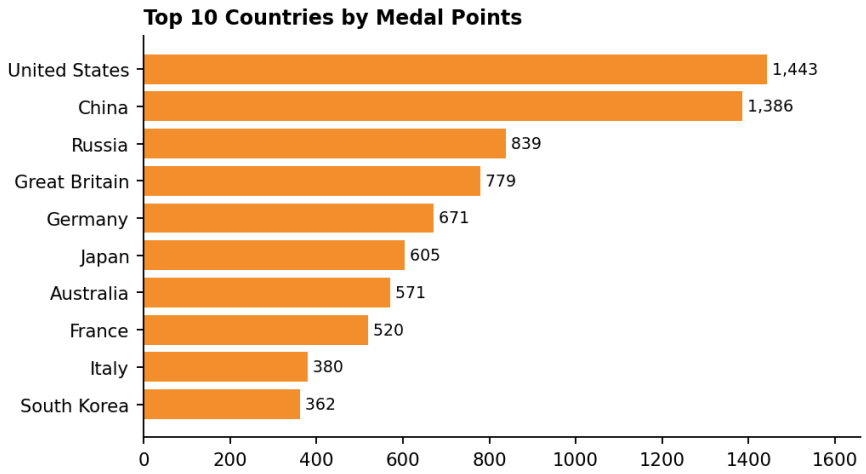


Figure: Worksheet 5: Top 10 Countries by Medal Points

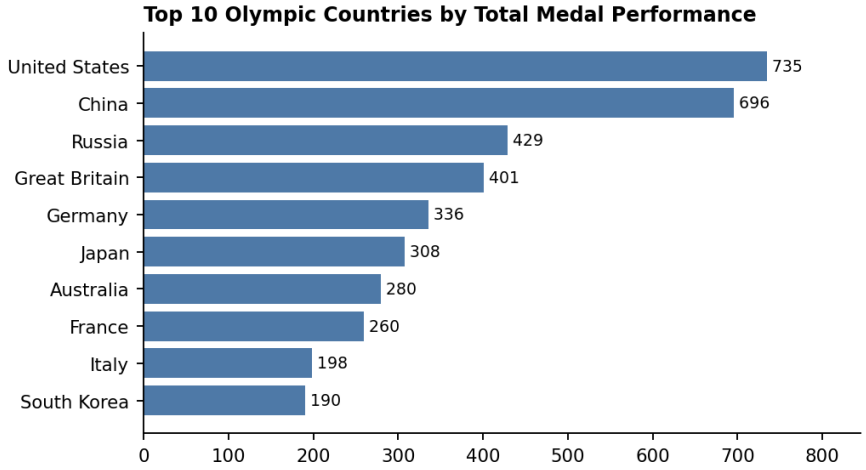


Figure: Worksheet 6: Top 10 Olympic Countries by Total Medal Performance

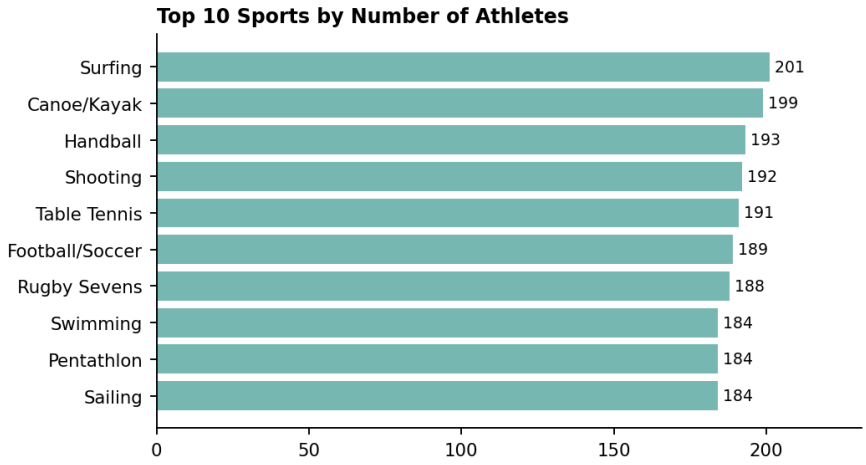


Figure: Worksheet 7: Top 10 Sports by Number of Athletes

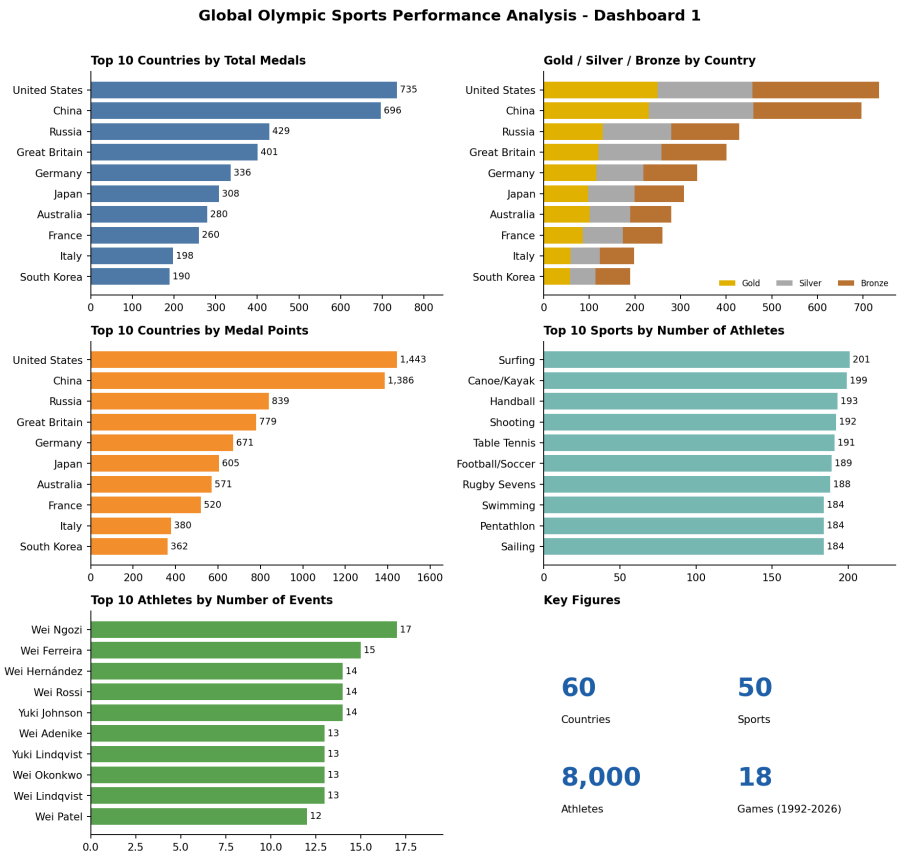


Figure: Dashboard 1 – Global Olympic Sports Performance Analysis (also Story 1, story point 1)

Key insights

#	Key insight	Evidence
1	United States and China lead by a wide margin	735 and 696 total medals; 23.8% of all 6,025 medal-result medals
2	Gold race is close	Gold: US 250, China 230; medal points US 1,443, China 1,386
3	Large gap to the next group	Russia 429, Great Britain 401, Germany 336
4	Participation is evenly spread	Surfing 201, Canoe/Kayak 199, Handball 193; Luge lowest at 100
5	Europe leads by region	2,822 of 6,025 medals (46.8%)
6	Gender balanced athlete records	4,179 male (52.2%), 3,821 female (47.8%)

8. ADVANTAGES & DISADVANTAGES

Advantages

- Converts 8,988 raw records into clear visuals.
- Interactive filters let users compare and drill into results.
- Consistent medal measures across all views.
- Reusable for new Games.
- Easy for non-technical users.

Disadvantages

- Four record types need dataset_type filters.
- Repeated athlete names need care (COUNTD of athlete_id is used).
- Static data with no live feed.
- Limited to 60 countries and 50 sports.

9. CONCLUSION

A well-prepared dataset and a focused set of visuals explain Olympic performance clearly: the United States and China lead, participation is spread across sports, and the dashboard and story make the findings easy to share.

10. FUTURE SCOPE

- Add a world map and a trend line across Games years.
- Add efficiency (medals per athlete sent) and GDP tier / investment analysis.
- Connect Tableau live to Snowflake for automatic refresh.
- Analyse athlete age, BMI and training hours versus medals.

11. APPENDIX

Source Code

Snowflake scripts: warehouse, database and schema setup, file format and stage, 56-column table, COPY INTO load and the five analysis queries (full scripts are in the Snowflake Code & Solution document).

```
CREATE WAREHOUSE OLYMPIC_WH WITH WAREHOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE;
CREATE DATABASE OLYMPIC_DB;  CREATE SCHEMA OLYMPIC_DB.SPORTS;

COPY INTO OLYMPIC_EVENTS FROM @OLYMPIC_STAGE/athelete_events.csv.gz
FILE_FORMAT = (FORMAT_NAME = CSV_FF) ON_ERROR = 'ABORT_STATEMENT';

SELECT country, SUM(total_medals) AS total_medals FROM olympic_events
WHERE dataset_type='medal_results' GROUP BY country ORDER BY total_medals DESC LIMIT 10;
```

Dataset Link

Item	Details
Dataset	athelete_events.csv (8,988 rows, 56 columns; Olympic data 1992–2026)
Dataset link	To be added by the team

GitHub & Project Demo Link

Item	Details
Workbooks	Global-Olympic-Sports-Performance-Analysis-Dashboard.twbx; Global-Olympic-Sports-Performance-Analysis-Story.twbx
GitHub link	To be added by the team
Project demo link	To be added by the team